

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	Course Title:	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	Assessment:	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	CO1: <i>Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)</i>		
Student Name:	Sameer Ahmed G	Reg. No.:	192421154
Section / Batch:	B Tech IT	Date:	27-07-2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.

Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.

For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.

No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
GRAND TOTAL:			100 Marks

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

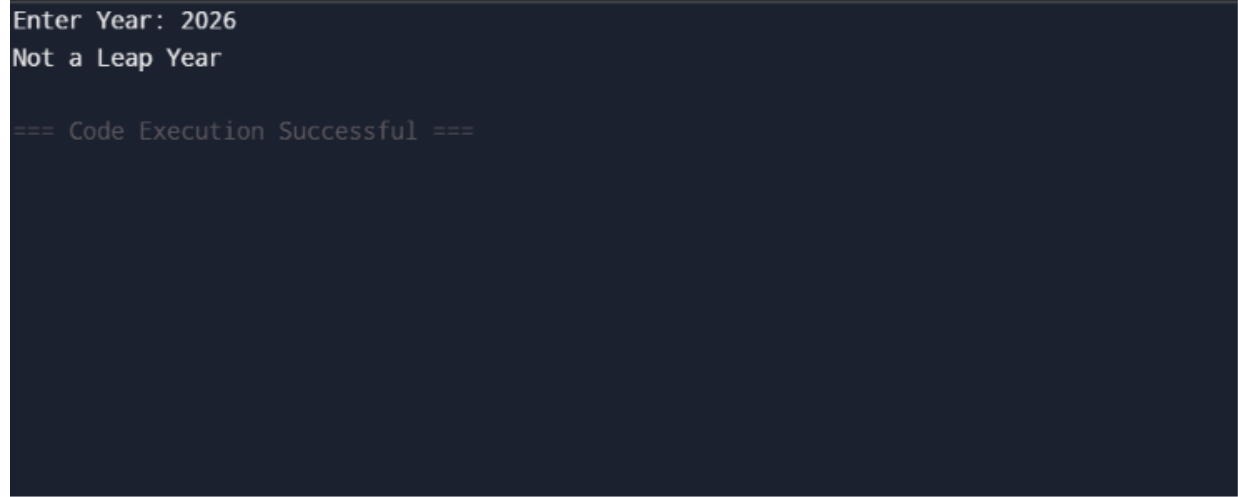
```
START

INPUT year

IF(year % 400 == 0 OR (year % 4 == 0 AND year % 100 != 0))
{
    DISPLAY "Leap Year"
}
ELSE
{
    DISPLAY "Not a Leap Year"
}

STOP
```

Answer / Program with output:



```
Enter Year: 2026
Not a Leap Year

=== Code Execution Successful ===
```

Q2. (Level 1 – Program Writing) [20 Marks]

```
START

INPUT string

FOR(i = length-1; i >= 0; i--)
{
    DISPLAY string[i]
}

STOP
```

Answer / Program with output:

```
Enter String: Sameer  
reemaS  
=== Code Execution Successful ===
```

Q3. (Level 1 – Program Writing) [20 Marks]

```
START  
  
INPUT n  
  
fact = 1  
  
FOR(i = 1; i <= n; i++)  
{  
    fact = fact * i  
}  
  
DISPLAY fact  
  
STOP
```

Answer / Program with output:

```
Enter Number: 5
Factorial = 120

=== Code Execution Successful ===
```

Q4. (Level 1 – Program Writing) [20 Marks]

```
START

INPUT rows, columns

INPUT Matrix1
INPUT Matrix2

FOR(i = 0; i < rows; i++)
{
    FOR(j = 0; j < columns; j++)
    {
        sum[i][j] = matrix1[i][j] + matrix2[i][j]
    }
}

DISPLAY sum

STOP
```

Answer / Program with output:

```

Enter Rows: 2
Enter Columns: 2
Enter First Matrix:
1
2
3
4
Enter Second Matrix:
1
12
3
4
Sum Matrix:
2 14
6 8

=== Code Execution Successful ===

```

RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I Sameer Ahmed G (192421154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sameer Ahmed G